

JetGraph: Graph Neural Networks for Quark/Gluon Jet Tagging

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Abstract

JetGraph is a compact high-energy physics and machine-learning study of quark/gluon jet tagging with particle-level jet constituents. Using the EnergyFlow `qg-jets` dataset, the project compares interpretable physics-inspired observables with graph neural networks based on EdgeConv. Observable-only baselines already provide strong discrimination, with gradient boosting reaching ROC AUC 0.8616. A first EdgeConv model trained on k-nearest-neighbour particle graphs reaches comparable performance, with the best hybrid node-feature setup reaching ROC AUC 0.8617 on the Pythia test set. Additional ablations study feature representation, graph connectivity, particle-identity information, and Pythia-vs-Herwig generator dependence.

1 Introduction

Quark/gluon tagging is a standard benchmark problem for studying jet substructure and machine-learning methods in collider physics. The task is challenging because jets are variable-size sets of particles, and because useful discrimination can arise from both global radiation patterns and local particle-level correlations.

JetGraph is designed as a reproducible mini-study of this problem. The analysis starts with simple physics observables, then moves to particle-level graph representations and a small graph neural network. This staged structure makes the comparison scientifically interpretable: the graph model is judged not only by its raw performance, but also by whether it improves on meaningful physics baselines and remains stable under controlled ablations.

2 Physics Motivation

Jets are collimated sprays of hadrons produced by energetic partons. Gluon jets typically radiate more than quark jets because gluons carry a larger color factor. As a result, gluon-initiated jets tend to have higher particle multiplicity, broader angular radiation, and different energy-sharing patterns. These properties motivate observables such as multiplicity, jet mass, and angular widths.

Graph neural networks offer a complementary representation. Instead of reducing the jet to a few global quantities, they operate on the particle cloud itself. This is close to the physical picture of a jet as a structured set of constituents, and is related to successful particle-cloud architectures such as ParticleNet [1].

3 Dataset

The study uses the EnergyFlow `qg-jets` dataset [2]. The primary sample contains 10,000 Pythia jets generated for quark/gluon classification. Each event is represented as a padded array of

particle constituents with columns

$$(p_T, \eta, \phi, \text{pid}).$$

Padded particles are removed by requiring $p_T > 0$. Herwig samples are used in the generator-robustness study.

4 Physics-Inspired Baseline Observables

The baseline model uses five global jet observables:

- particle multiplicity, excluding padded constituents;
- scalar constituent p_T sum;
- approximate jet mass from massless constituent four-vectors;
- p_T -weighted η width;
- p_T -weighted ϕ width.

These quantities encode the expected differences between quark and gluon radiation patterns. Logistic regression, random forest, and gradient boosting are trained on these observables.

5 Graph Construction

For the graph-based model, each jet is converted into a PyTorch Geometric `Data` object [3]. Nodes correspond to particles, and directed edges are built with k -nearest neighbours in the wrapped (η, ϕ) plane. Unless otherwise stated, $k = 8$.

Several node-feature modes are studied:

- **raw**: $p_T, \eta, \phi, \text{pid}$;
- **physics**: $\log p_T, p_T$ fraction, $\Delta\eta, \Delta\phi, \Delta R, \text{pid}$;
- **hybrid**: raw kinematics plus the relative physics features;
- no-pid variants where particle identity is removed.

The relative coordinates are computed with respect to the p_T -weighted jet axis, with $\Delta\phi$ wrapped to the usual angular interval.

6 EdgeConv Architecture

The first graph model is intentionally simple. It consists of two EdgeConv layers with ReLU activations, followed by global mean pooling and a small MLP classifier. The model is trained with cross-entropy loss for binary classification. This minimal architecture is sufficient for controlled experiments on feature representation and graph connectivity.

7 Results

7.1 Baseline Comparison

The observable-only baselines are already highly competitive, as shown in Table 1. Gradient boosting reaches ROC AUC 0.8616.

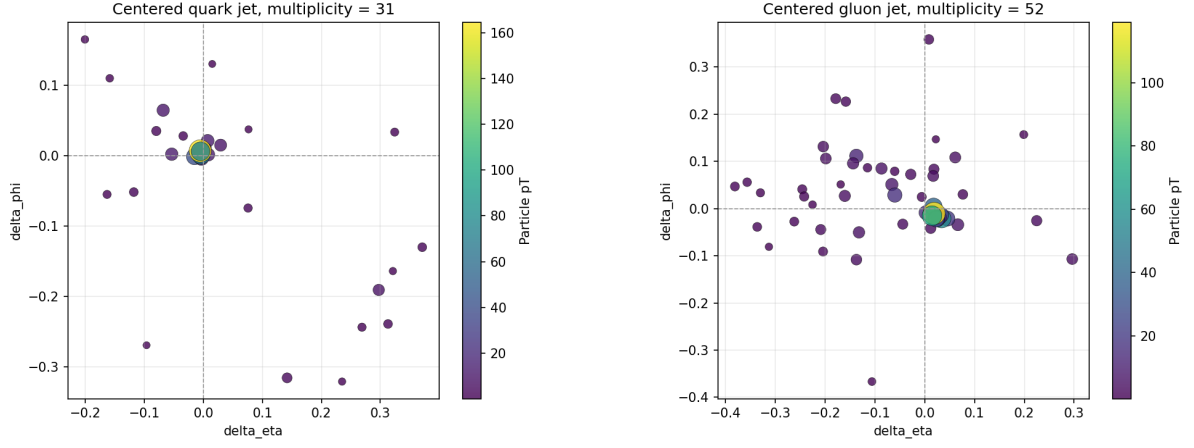


Figure 1: Centered quark and gluon jet displays in $(\Delta\eta, \Delta\phi)$ coordinates. Marker size is proportional to particle transverse momentum.

Table 1: Performance of physics-observable baseline classifiers.

Model	ROC AUC
Logistic regression	0.8579
Random forest	0.8451
Gradient boosting	0.8616

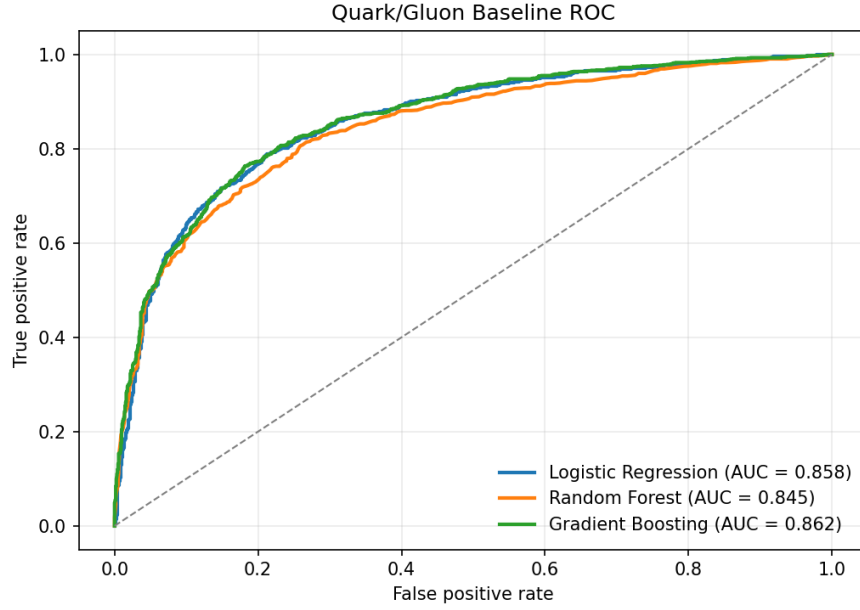


Figure 2: ROC curves for the physics-inspired baseline classifiers.

7.2 Feature Ablation

The EdgeConv model is sensitive to node-feature representation. Raw features already work well, physics-only relative features underperform, and hybrid features recover the best performance.

Table 2: EdgeConv node-feature ablation.

Feature mode	ROC AUC
Raw	0.8580
Physics	0.8102
Hybrid	0.8617

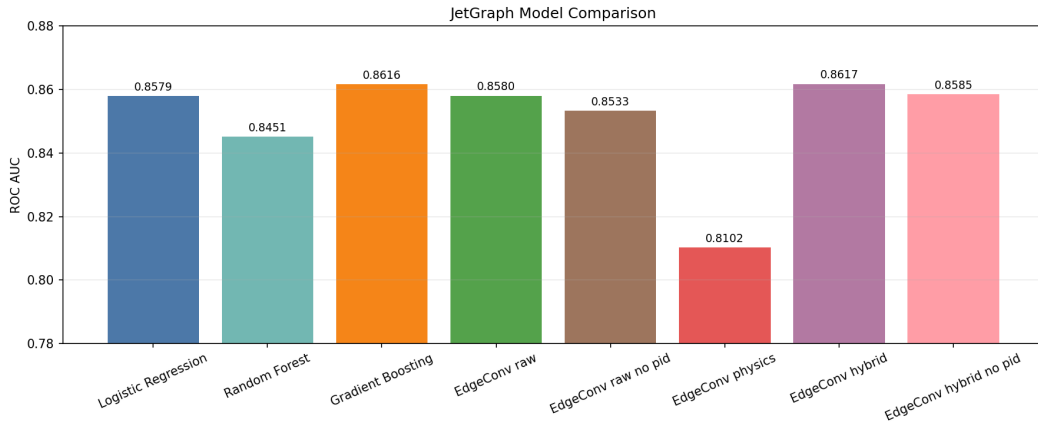


Figure 3: Summary of baseline and EdgeConv model comparisons.

7.3 k-Connectivity Study

Using hybrid node features, the graph connectivity parameter is scanned over $k = 4, 8, 12, 16$. The best AUC in this sweep is obtained at $k = 8$.

Table 3: Impact of the k-nearest-neighbour connectivity parameter.

k	Test accuracy	Test ROC AUC
4	0.7653	0.8525
8	0.7707	0.8579
12	0.7720	0.8511
16	0.7640	0.8455

7.4 pid/no-pid Study

Particle identity information provides a small but visible gain. Removing `pid` lowers the raw-feature AUC from 0.8580 to 0.8533 and the hybrid-feature AUC from 0.8617 to 0.8585.

7.5 Generator Robustness

The generator-robustness study trains on one generator and tests on both Pythia and Herwig [4, 5]. The results in Table 5 show a significant drop when a Pythia-trained model is evaluated on

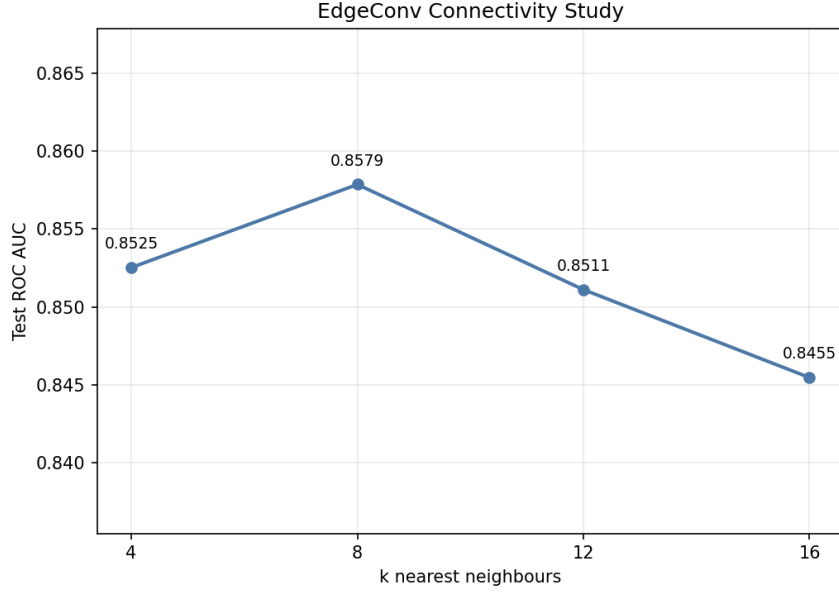


Figure 4: Test ROC AUC as a function of kNN graph connectivity.

Table 4: Effect of removing particle identity from node features.

Feature family	With pid	Without pid
Raw	0.8580	0.8533
Hybrid	0.8617	0.8585

Herwig.

Table 5: Pythia-vs-Herwig generator robustness.

Train generator	Test generator	Accuracy	ROC AUC
Pythia	Pythia	0.7707	0.8579
Pythia	Herwig	0.7147	0.7835
Herwig	Pythia	0.7693	0.8464
Herwig	Herwig	0.7053	0.7955

8 Discussion

The study demonstrates that simple physically motivated observables remain a strong baseline for quark/gluon tagging. The EdgeConv graph model reaches similar performance only after careful feature design. In particular, removing absolute kinematic information in the physics-only representation degrades the result, while hybrid features combining absolute and relative coordinates perform best.

The graph-connectivity scan suggests that moderately local neighbourhoods are useful, but denser graphs do not automatically improve performance. The generator study is the main cautionary result: strong in-domain performance does not imply robust physics learning across different Monte Carlo generators.

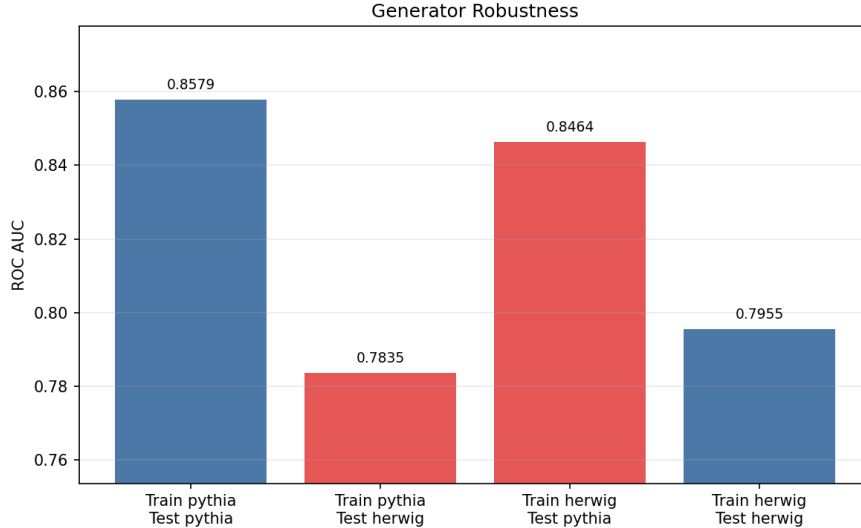


Figure 5: Cross-generator ROC AUC comparison for EdgeConv models trained on Pythia or Herwig.

9 Conclusion and Future Work

JetGraph provides a concise, reproducible mini-study of graph learning for quark–gluon tagging. The best current EdgeConv result, ROC AUC 0.8617 with hybrid node features, is comparable to the best gradient-boosted observable baseline. This confirms both the strength of traditional physics-inspired variables and the potential of graph-based representations.

Future work should study larger samples, repeated random seeds, train-set-only feature normalization, dynamic graph recomputation, and ParticleNet-like architectures. The most important next step is to improve Pythia-vs-Herwig robustness, for example with mixed-generator training or domain-adversarial objectives.

References

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